

Lovely Professional University, Punjab

Course Code	Course Title	Lectures	Tutorials	Practicals	Credits	
ECE279	BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY	0	0	2	1	
Course Weightage Course Focus	ATT: 5 CAP: 45 ETP: 50 EMPLOYABILITY,SKILL DEVELOPMENT	Exam Category: X6: Mid Term Exam: Not Applicable – End Term Exam: Practical				

Course Outcomes :Through this course students should be able to

CO1 :: assemble various measuring instruments and electronic components on the breadboard and create circuit connections.

CO2 :: execute the functionality test of various digital & analogue ICs

CO3 :: use basic measuring instruments and component specific ratings related to electronics engineering

CO4 :: design applications and analyze combinational and sequential circuits

CO5 :: communicate and contribute effectively as an individual and as a member of a team

CO6 :: build applications with at least one sensor module by programming Microcontroller board

	Reference Books (R)		
Sr No	Title	Author	Publisher Name
R-1	FUNDAMENTALS OF ELECTRICAL ENGINEERING AND ELECTRONICS	B.L.THERAJA	S Chand Publishing
R-2	DIGITAL DESIGN PRINCIPLES AND PRACTICES PEARSON	JOHN F. WAKERLY	PEARSON
R-3	INTERNET OF THINGS	RAJ KAMAL	MCGRAW HILL EDUCATION
R-4	DIGITAL ELECTRONICS, PRINCIPLES, DEVICES AND APPLICATIONS	ANIL K. MAINI	JOHN WILEY & SONS
R-5	DIGITAL LOGIC AND COMPUTER DESIGN	M. MORRIS MANO	PEARSON

Virtual Labs (VL)		
Sr No	(VL) (only if relevant to the course)	Salient Features
VL-1	https://virtual-labs.github.io/exp-kirchhoff-law-iitr/pretest.html	KVL
VL-2	https://da-iitb.vlabs.ac.in/exp/washin-machine-control/	Washing machine controller using logic gates

An instruction plan is only a tentative plan. The teacher may make some changes in his/her teaching plan. The students are advised to use syllabus for preparation of all examinations. The students are expected to keep themselves updated on the contemporary issues related to the course. Upto 20% of the questions in any examination/Academic tasks can be asked from such issues even if not explicitly mentioned in the instruction plan.

Scheme for CA:

CA Category of this Course Code is:A0303 (3 out of 3)

Component	Weightage (%)	Mapped CO(s)
Performance/Job Evaluation/Execution 1	33.33	CO1, CO2, CO3
Performance/Job Evaluation/Execution 2	33.33	CO1, CO2, CO3
Test- Situation based problem solving	33.33	CO1, CO2, CO3, CO4

Details of Academic Task(s)

Academic Task	Objective	Detail of Academic Task	Nature of Academic Task (group/individuals)	Academic Task Mode	Marks	Allottment / submission Week
Performance/Job Evaluation/Execution 1	Evaluation of workbook (Practical 1, 2 and 3)	Faculty will evaluate workbook of students for practical 1, 2 and 3. 50 Marks for each practical, Total 150 marks.	Individual	Offline	150	3 / 4
Performance/Job Evaluation/Execution 2	Evaluation of workbook (Practical 4, 5 and 6)	Faculty will evaluate workbook of students for practical 4, 5 and 6. 50 Marks for each practical, Total 150 marks.	Individual	Offline	150	8 / 9
Test- Situation based problem solving	Pictionary Based Activity	Faculty will conduct Pictionary Based Activity type assessment, based on Practical 1 to Practical 4) Total Question 15 and maximum marks 60 This will evaluate practical conducting ability of the student.	Individual	Offline	60	4 / 7

Detailed Plan For Practicals

Practical No	Broad topic	Subtopic	Other Readings	Learning Outcomes
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Practical 1	The Electronics Expedition	Familiarization of electronics components, trainer module, input sources and display devices related to electronic circuits		1. Observe the various electrical elements of the electric circuit. 2. Observe the different instruments which are usually needed for measurement. 3. Understand the use of the instrument and components in the experiment 4. Make students aware of course outcomes and its map with program outcomes.
Practical 2	KVL Voltage Hunt	Validation and analysis of Kirchhoff's Voltage Law (KVL)	VL-1	1. Identify values of resistances using color coding scheme. 2. Use ammeter and voltmeter 3. Verify KVL theorem. 4. Use breadboard and make connections
Practical 3	Diode Voltage Quest	Analyze V-I characteristics of PN junction diode		1. Measure diode current and voltage in forward biased condition. 2. Test the diode for functioning. 3. Identify different sources of error.
Practical 4	Performance/Job Evaluation/Execution 1			
Practical 5	Universal Gate Adventure	Implementation and validation of logic gates and their design using universal gates	VL-2	1. Verify the functionality of Logic Gates 2. Implement Basic Logic gates using Universal gates and verify their function.
Practical 6	Adder Chronicles	Design half adder, full adder and validate the combinational logic	VL-3	1. Implement Full Adder (FA) using logic gates. 2. Verify the working of FA using logic gates. 3. Verify the functioning of IC-7486, 7408 and 7432
Practical 7	Test- Situation based problem solving			
Practical 8	Flip-Flop Warriors: NAND Edition	Design and validate JK flip-flop with NAND gates	VL-3	1. Implement JK flip flop using NAND gates and verify of its functionality. 2. Implement T flip flop using NAND gates and verify its functionality.
Practical 9	Performance/Job Evaluation/Execution 2			
Practical 10	List of Projects	Design and implementation of 2-bit ALU performing arithmetic and logical operations with display		1. Create applications for combinational and sequential circuits. 2. Build innovative solutions using one sensor and microcontroller board.
Practical 11	List of Projects	Design and implementation of Electronics Frills using shift register or counters		1. Create applications for combinational and sequential circuits. 2. Build innovative solutions using one sensor and microcontroller board.
Practical 12	List of Projects	Design and implementation of 2-dimensional memory controller using decoders		1. Create applications for combinational and sequential circuits. 2. Build innovative solutions using one sensor and microcontroller board.

Practical 13	List of Projects	Design and implementation of digital clock with dual digit hour, minute and second display		1. Create applications for combinational and sequential circuits. 2. Build innovative solutions using one sensor and microcontroller board.
Practical 14	List of Projects	Design and implementation of encryption-decryption keys for error free communication		1. Create applications for combinational and sequential circuits. 2. Build innovative solutions using one sensor and microcontroller board.
	SPILL OVER			
Practical 15	Spill Over			